

Operations & Logistics Intelligence Project

Supply Chain Graduation Project (2022-2024)

Instructor

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SUPPLY CHAIN
LOGISTICS

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1. Project Overview

Logistics Operational Analysis (2022-2024)

Comprehensive analysis of a realistic logistics operational database spanning **three years**.

Rising operating **costs** and **declining** operational **efficiency** was the critical warning sign

360,000+

Raw Records

14

Interconnected Tables

120 / 200

Trucks / Customers

Strategic Goals & Objectives



Optimize Fleet Performance



Reduce Operational Costs



Enhance Service Reliability

Data Pruning Process

We cleaned the **trips** table by filtering out missing values in key identifier fields (**driver_id**, **truck_id**, **trailer_id**). This ensures statistical integrity and unbiased profitability metrics for our analysis.

BEFORE: Raw Data

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 85410 entries, 0 to 85409
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   trip_id               85410 non-null  object
1   load_id               85410 non-null  object
2   driver_id             83696 non-null  object
3   truck_id              83738 non-null  object
4   trailer_id            83730 non-null  object
5   dispatch_date         85410 non-null  object
6   actual_distance_miles  85410 non-null  int64
7   actual_duration_hours  85410 non-null  float64
8   fuel_gallons_used     85410 non-null  float64
9   average_mpg           85410 non-null  float64
10  idle_time_hours       85410 non-null  float64
11  trip_status           85410 non-null  object
dtypes: float64(4), int64(1), object(7)
memory usage: 7.8+ MB
```



AFTER: Cleaned Dataset

```
Index: 80458 entries, 0 to 85409
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   trip_id               80458 non-null  object
1   load_id               80458 non-null  object
2   driver_id             80458 non-null  object
3   truck_id              80458 non-null  object
4   trailer_id            80458 non-null  object
5   dispatch_date         80458 non-null  datetime64[ns]
6   actual_distance_miles  80458 non-null  int64
7   actual_duration_hours  80458 non-null  float64
8   fuel_gallons_used     80458 non-null  float64
9   average_mpg           80458 non-null  float64
10  idle_time_hours       80458 non-null  float64
11  trip_status           80458 non-null  object
dtypes: datetime64[ns](1), float64(4), int64(1), object(6)
memory usage: 8.0+ MB
```

1 - Trips Table: Data cleaning

We performed **Data filtering** on the **trips** table to remove inconsistent records.

By filtering out **incomplete data**, we ensured the **Statistical Integrity** of our performance metrics, preventing biased results in profitability analysis.

We removed incomplete rows, resulting in **80,458 rows** of high-quality data and converted date columns to a standard format.

Data Cleaning Summary

- **Total Raw Records:** 85,410
- **Records Removed:** 4,952
- **Data Loss Rate:** 5.80%
- **Final Dataset:** 80,458 rows

5.8% Gap Discovered in trip records.

Recommendation: Fix logging software to make identifier fields mandatory.

"In a real trucking company, you cannot have a trip without a driver or a truck. It's like a ghost trip".

This prevents data loss and ensures 100% accurate future reporting without manual cleaning.

🔍 فحص القيم المفقودة في جدول الرحلات (Trips):

```
trip_id      0
load_id      0
driver_id    1714
truck_id     1672
trailer_id   1680
dispatch_date 0
actual_distance_miles 0
actual_duration_hours 0
fuel_gallons_used 0
average_mpg  0
idle_time_hours 0
trip_status  0
dtype: int64
```

Data Synchronization: Maintaining Integrity Across Tables

After cleaning the **Trips** table, other tables (Fuel, Delivery Events, Safety, Loads) were synchronized to ensure analysis of valid trips only.

Why this is important?

- 1. Consistency:** Ensuring **trip_id** exists in all tables; no fuel costs for a deleted trip!
- 2. Accurate KPIs:** Keeping deleted records leads to errors in "Total Cost" and profit analysis.
- 3. Prevent Orphaned Records:** Prevents "ghost" data that doesn't belong to any real trip.



Clean Trip IDs were used as a filter for all linked tables.

How it works: If a trip ID is deleted from the main table, it is automatically removed from all child tables.

Comprehensive Data Cleaning & Synchronization

Drivers & Trips Synchronization

Driver records were cleaned and synchronized.

Number of drivers before filtering: **150**

Remaining drivers (who have trips): **124**

Fuel & Loads Refinement

- Fuel table cleaned & dates converted.
- Loads table texts cleaned; revenue calculated via load ID.

Fleet & Infrastructure Tables

The following specialized tables were thoroughly cleaned and verified:

- **Assets:** Truck table, Trailers
- **Operations:** Maintenance Records, Routes, Facilities
- **Safety:** Safety Incidents Profiling



All supporting operational tables have been fully cleaned and synchronized.

Categorizing Delivery Punctuality

Arrival Status Methodology

In the `Delivery Events` table, we identified negative time differences, which represent early arrivals.

We created a new column called `arrival_status` to categorize each event:

- **Early:** Negative time difference.
- **Late:** Positive time difference.

This transforms raw numbers into clear, visual insights about driver punctuality.

Data Transformation Example

	event_type	arrival_delay_minutes	arrival_status	on_time_flag
0	Pickup	178.931970	Late	False
1	Delivery	-100.562935	Early	True
2	Pickup	-22.556526	Early	True
3	Delivery	222.327164	Late	False
4	Pickup	-31.963839	Early	True

Note: The `on_time_flag` is derived directly from the `arrival_status` to simplify KPI reporting.

Performance Scorecard

Reporting Period: 2022 – 2024



73.4%

OTD Rate
On-Time Delivery



68.1%

OTF Rate
On-Time In-Full



427

Loads/Customer
Average



\$298.6M

Total Revenue
Across All Routes



200

Total Customers
Active Accounts



\$1.49M

Total Rev/Customer
Per Account



\$3.5K

Avg Rev/Customer
Per Account



\$91.8M

Total Fuel Cost
65% Margin (Fuel Only)



\$4.3M

Maint. Cost (3yr)
\$1.4M per Year



83.0%

Fleet Utilization
Trucks Active

Performance metrics are monitored monthly to ensure operational excellence across all logistical channels.



Supply Chain & Logistic Analysis

\$298.6M
Total Revenue

65.02%
Route_Profit_Margin...

\$91.8M
Total Fuel Cost

\$4.3M
Total Maintenance Costs

83.04%
Truck Utilization Rate

Overview

Customer
Performance

Driver
Performance

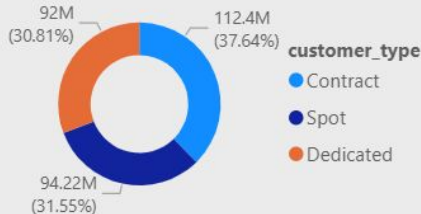
Fleet Utilization

Shipment &
Trip...

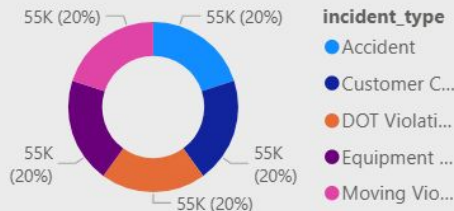
Route
Profitability

Cost
Management

Total Revenue by customer_type



Downtime by incident_type



Year

Desel...

2022

2023

2024

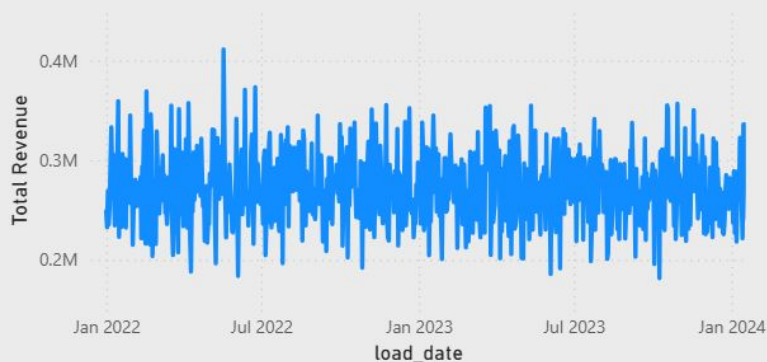
Total Maintenance Costs by make



Top 10 Revenue By Route



Total Revenue by load_date



Overview Dashboard: Strategic Insights & Recommendations

\$298.6M

TOTAL REVENUE

\$91.8M

TOTAL FUEL COST

65%

PROFIT MARGIN

\$4.3M

MAINTENANCE (3YR)

83%

UTILIZATION RATE

1. Fuel & Routing

Critical Area: High Costs

- Improve route optimization
- Reduce driver idle time
- Monitor consumption/truck
- Replace low MPG units
- Driver efficiency training

2. Asset Maintenance

Critical Area: High Expense

- Shift to preventive model
- Analyze equipment age
- Track downtime frequency
- Replace high-cost trucks

3. Operational Risk

Critical Area: Incidents

- Incident prevention program
- Safety & DOT compliance
- Defensive driving training
- Regular inspections

Strategic Actions: Improve Fuel Efficiency • Expand Profitable Routes • Enhance Predictive Maintenance • Secure Long-Term Contracts

Driver Performance Analysis Report

1. Fleet Productivity & Activity

Metric: Avg Miles/Driver: **26.81K** | Drivers: **124**

Insight: High workforce utilization is achieved, but pressures maintenance schedules.

Rec: Monitor HOS (**Hours of Service**) and distribute long-haul trips to prevent burnout.

2. Fuel Efficiency & Environment

Metric: Avg MPG: **6.50** | Avg Fuel Cost: **\$612.2K**

Insight: Marginal MPG improvements can save hundreds of thousands of dollars.

Rec: Launch "Eco-Driving" Training to target **7.0** MPG (7% reduction).

3. Operational Bottlenecks

Metric: Idle: 7.01 Hrs | Detention: 91.56 Min

Insight: Non-productive idling and waiting consume fuel and lower satisfaction.

Rec: Use Auto Engine Management and renegotiate detention terms.

4. Safety & Compliance

Metric: Accidents: **170** | Claims: **\$1.05M**

Insight: Accident rate exceeds 1 per driver; claims have crossed the \$1M mark.

Rec: Pivot to Safety Scorecard incentives to reduce claim liabilities.

This performance summary identifies critical areas for safety, efficiency, and productivity gains.

Shipment & Trip Performance Comprehensive Report

1. Fleet Financial Productivity

Metric: Avg Rev/Load: **\$3.07K**

Insight: Tight margins due to high fuel costs.

Rec: Review "Under-Performing" routes to push avg **>\$3.5K**.

2. Payload Capacity Utilization

Metric: Avg Load Weight: **27.48K lbs**

Insight: Moderate capacity, significant "Empty Space" remains. -still have space-

Rec: Implement Load Optimization for **35K-40K lbs range**.

3. Operational Punctuality Gap

Metric: Avg Arrival Delay: 105.40 Mins

Insight: ~1h 45m delay drives detention costs and frustration.

Rec: Audit Dispatch Scheduling for better estimations.

4. Delivery Cycle Speed

Metric: Avg Delivery Time: 1.00 Day

Insight: High velocity in short-to-medium haul segment.

Rec: Maintain 1-day standard as a Competitive Advantage.

5. Service Reliability

Metric: Rejected Loads: 0

Insight: 100% Commitment Rate is a rare, valuable KPI.

Rec: Use "100% Reliability" as a lead marketing message.

6. Workforce Engagement

Metric: Total Number of Trips: (Insert Total)

Insight: High volume handled by 124 drivers daily.

Rec: Balance volume with **83%** utilization to avoid wear.

Fleet Utilization Executive Report

1. Fleet Productivity & Capacity

Metric: Truck Utilization Rate: **0.83 (83%)**

Insight: Minimal idle assets, strong demand.

Rec: Maintain 17% buffer for spikes.

2. Mechanical Performance & Endurance

Metric: Avg Miles/Truck: **36.15K**

Insight: High stamina but accelerates wear.

Rec: Use data for Automated Maintenance Alerts.

3. Asset Revenue Generation

Metric: Revenue per Truck: **\$2.80M**

Insight: Strong ROI per vehicle.

Rec: Replace older units with lower revenue.

4. Maintenance Cost Efficiency

Metric: Cost per Mile: **\$0.04** Rising from \$0.035 = **14% increase in 4 years.**

Insight: Exceptionally efficient routine repairs.

Rec: Benchmark against industry standards. Aging fleet signal replacement planning should begin now for trucks >8 years old.

Operational excellence through proactive maintenance and downtime reduction.

Strategic Operational Insights & Action Plan

1. Profitability Crisis

Insight: Total Fuel Costs (**\$91.8M**) are severely outpacing Route Revenue (**\$298.6M**).

Action: Implement **Dynamic Fuel Surcharges** and renegotiate high-distance rates.

2. Operational Leaks

Insight: **Idling (7.0 hrs)** and **Detention (91.6 min)** drive fuel waste.

Action: Launch **"Zero-Idling" Incentives** and optimize scheduling.

4. Financial Risk

Insight: Safety claims have exceeded **\$1M**, eroding net profitability.

Action: Conduct **Regional Safety Workshops** to reduce claim costs.

Critical operational findings requiring immediate execution of the action plan.

Cost Management Analysis Report

1. Fuel Expenditure Domination

Metric: Total Fuel Cost: **\$91.83M** | Avg/Unit: **\$3.90**

Insight: Primary expense 1% reduction saves nearly **\$1M** annually.

Recommendation: Centralized Fuel Procurement & GPS-based Route Optimization.

2. Maintenance Efficiency

Metric: Total Maintenance Cost: **\$4.33M**

Insight: Well-managed at **4.7%** of fuel spend cross-reference with availability.

Recommendation: Shift focus to **Preventative Maintenance** for older units.

3. Safety & Incident Liability

Metric: Incident Costs: **\$1M**

Insight: "Lost capital" eroding net profit margin directly.

Recommendation: Reinvest in **ADAS -Advanced Driver Assistance Systems-** & training; 25% reduction adds **\$250K** to bottom line.

Critical areas for financial and operational optimization across the fleet.

Customer Performance Dashboard

Total Revenue
\$205.2M

Total Customers
200

Loads per...
294

Total Revenue...
\$1.03M

Avg. Revenue
\$3.49K

1. Automotive Sector Dominance

Insight: Automotive customers generate the largest share of revenue.

Solution:

- Focus more on Automotive clients.
- Expand services for this sector.

2. Contract Customer Value

Insight: Contract customers provide stable and high revenue.

Solution:

- Increase long-term contracts.
- Improve customer relationships.

3. Revenue Dependence Risk

Insight: A small number of customers generate most of the revenue.

Solution:

- Attract new customers.
- Diversify the customer base.

4. Low Customer Density Sectors

Insight: General and Consumer Goods sectors have lower customer numbers.

Solution:

- Increase marketing efforts.
- Offer competitive pricing.

Strategic growth initiatives focusing on high-revenue sectors and stable contract partnerships.

Thank You